

Statistics 250, Section 003, Quiz 1 (150 Points)

February 17, 1999, Dr. Jürgen Symanzik

Your Name: _____

Question 1: Numbers and Graphs (70 Points)

The following questions are based on the data for the “Weider Empire” from USA Today. **When you write down your answers, you should indicate which values are assigned to the variables you use (e.g., $n = 20$, $x_{(15)} = 20,000$) and which formula you use. This will guarantee you at least partial credit in case that your final result is incorrect.**

1. Determine the mean magazine circulation for the “Weider” empire. (10 Points)
2. Determine the median magazine circulation for the “Weider” empire. (10 Points)
3. Assume you are the press speaker of the “Smith” magazine group that also has 10 magazines with a mean circulation of 310,000 and a median circulation of 300,000. Which number(s) would you report when comparing your group with the “Weider” empire. Explain. (10 Points)

4. Calculate the range of the magazine circulation for the “Weider” empire. (**5 Points**)
5. Construct a stem-and-leaf display of the magazine circulation. (**15 Points**)
6. Are you happy with the data provided by USA Today? Isn’t some important information missing? (**10 Points**)
7. Do you think that the 1.1 million circulation for “Shape” is the absolute truth? Think of possible manipulations (hint: “TV Guide” published 4 collector’s covers of Seinfeld in their May 9–15, 1998, issue — how might this effect the circulation?). (**10 Points**)

Question 2: Newspaper Graphics (**40 Points**)

Look at the following graphics which have been republished in Howard Wainer (1997) “Visual Revelations”, Copernicus.

1. This graphic originally has been published in the *New York Times* in 1978. It contains at least 2 *graphos* (the analogue of textual *typos*). Which? (**15 Points**)

2. This graphic originally has been published in the *Washington Post*, Jan. 11, 1979.

Is there something wrong with it? If so, what?

Would you agree that the income growth is almost linear (i.e., data points fall close to a straight line) for both Physicians and Workers, however the slope for Physicians is much larger than the slope for Workers?

If you think the graphic is OK, you can stop here and just write **OK**. If you can think of a better representation of the same data, then go ahead and redraw the graphic. (**25 Points**)

Question 3: Mosaic Plots (**40 Points**)

Look at the Mosaic Plot of the Titanic data again.

1. Mark the following regions, using the letters “A”, “B”, and “C”:
“A”: Second Class, Child, Male, Not Survived.
“B”: Third Class, Child, Female, Not Survived.
“C”: Crew, Adult, Female, Survived.
(15 Points)
2. Indicate which of these statements are correct and which are incorrect:
(i) A very large percentage of First Class, Adult, Female survived.
(ii) Among the Adult, Male passengers, the largest percentage that survived was in First Class.
(iii) There were more Male than Female children on board of the Titanic.
(iv) There were more Third Class than First Class passengers.
(v) There were more Male Third Class passengers than Male First Class and Male Second Class passengers together.
(25 Points)